

PLATE 01 — ADEN NURLAYA (SMAN 1 SUMEDANG)

MEXT SCHOLARSHIP CANDIDATE • GRADE 12

Aden Nurlaya

Dedicated Mathematics & Tech Enthusiast | Student Researcher

Senior student at SMA Negeri 1 Sumedang, West Java. Uniting the analytical discipline of applied mathematics, software engineering, and competitive martial arts to build reliable, high-integrity digital systems for public administration.

INSTITUTION	PRIMARY TARGET	ATHLETIC RANK
SMAN 1 Sumedang	**MEXT Japan & IPDN**	**O2SN Karate Gold Medalist**

{ CORE PHILOSOPHY / I AM }

"I believe that complex mathematical reasoning, software development, and athletic grit are not separate disciplines, but three expressions of the same commitment: pursuing rigorous self-mastery to create trustworthy, ethical solutions for the community."

— Aden Nurlaya

Sumedang, West Java • 2026

Curated Work Samples

Class XII Framework: Problem → Process → Product → Impact → Reflection

CASE 01 • SCIENTIFIC INQUIRY

The Impact of FOMO on Deviant Behavior Among High School Students

May – June 2025

01 / THE PROBLEM

Adolescent peer circles are increasingly hijacked by social-media induced Fear of Missing Out (FOMO). Students regularly succumb to negative conformity and hazardous behaviors simply to maintain superficial validation within their peer groups.

02 / THE RESEARCH PROCESS

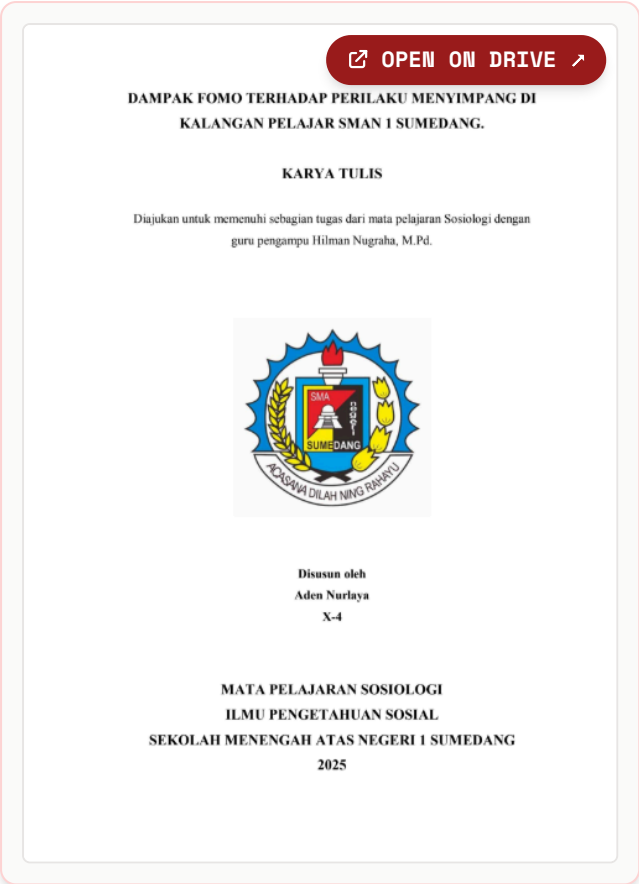
Formulated empirical hypotheses and conducted in-depth qualitative one-on-one interviews across varied student social clusters at SMAN 1 Sumedang, transcribing and coding narratives to uncover the root psychosocial motivations of teen deviance.

03 / THE PRODUCT & CONCRETE IMPACT

Produced a formal Scientific Research Paper (*Karya Tulis Ilmiah*). Distinctively, the primary recommendations were addressed directly to the **family and household environment** rather than conventional punitive school measures, identifying parental awareness as the foremost proactive safeguard.

> **04 / PERSONAL REFLECTION**
> *"FOMO misdirected into peer-pleasing can derail adolescent potential. My deepest takeaway is that FOMO must be handled with conscious intentionality: channel the drive to participate solely toward deliberate, positive growth rather than impulsive trends."*

05 / RESEARCH PUBLICATION & FULL MANUSCRIPT ACCESS



↓ Tap cover to read full paper on Google Drive ↗

CASE 02 • SOFTWARE & HARDWARE INTEGRATION

DiPay – Contactless Mobile Payment Ecosystem (NFC Tap-to-Pay)

`Kotlin • Node.js • NFC HCE`

01 / THE PROBLEM

Physical cash dependence in high-density campus environments leads to bottleneck queues, transaction delays, and reconciliation inaccuracies across student cafeterias and school co-ops.

02 / ENGINEERING PROCESS

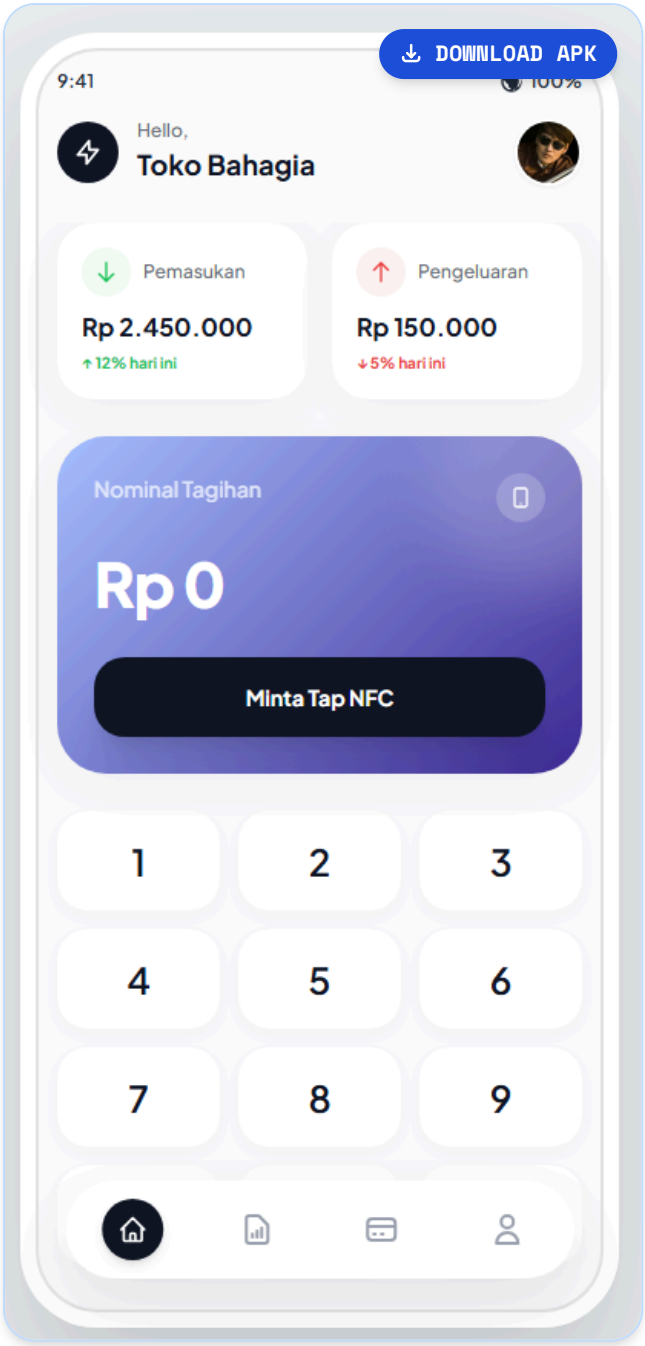
Engineered an Android mobile application leveraging NFC Host Card Emulation (HCE) to mimic contactless smart cards, connected to a dedicated Node.js backend server verifying cryptographic tokens and account balances in sub-second roundtrips.

03 / PRODUCT & PRACTICAL IMPACT

Successfully deployed the complete executable APK (`DiPay.apk`) with backend integration, validating an ultra-fast, offline-capable tap-to-pay transaction flow suitable for localized academic ecosystems.

> **04 / PERSONAL REFLECTION**
> *"Speed and security in IoT protocols must be architected simultaneously. Building DiPay proved that convenience must never compromise data integrity."*

05 / LIVE APP VISUAL EVIDENCE



↓ Tap image to download DiPay.apk (Android)

CASE 03 • MOBILE APP & DATA ANALYTICS

KeuanganKu — Smart Budget Allocation & Habit Analytics

`Android Studio • Java/Kotlin`

01 / THE PROBLEM

Low financial literacy among high schoolers manifests as irregular bookkeeping, inability to segregate essential allocations from impulsive wishes, and persistent pocket-money deficits.

02 / PRODUCT DEVELOPMENT

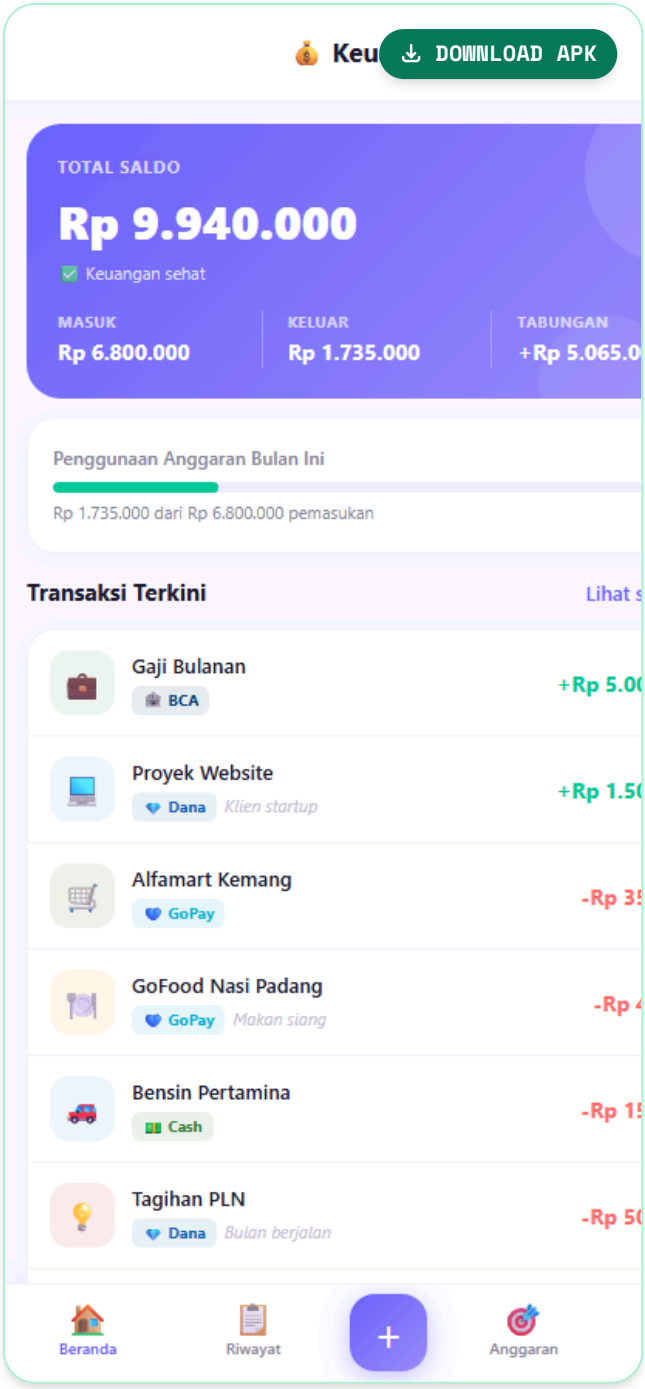
Architected a friction-free mobile experience with custom categorization algorithms, monthly budget thresholds, and clear graphical visualizations displaying spending velocity.

03 / DEMONSTRATED IMPACT

Packaged into `KeuanganKuApp`, providing a clean, self-contained personal accounting dashboard that empowers young individuals to establish disciplined savings habits without requiring cloud synchronization.

> ### 04 / PERSONAL REFLECTION
> *"Software efficacy is measured not by how complicated the codebase is, but by how reliably it shifts human behavior toward greater discipline and clarity."*

05 / LIVE APP VISUAL EVIDENCE



↓ Tap image to download KeuanganKu.apk (Android)

CASE 04 • EDGE AI & COMPUTER VISION

ShutterAI Camera — Real-Time On-Device Pose Guidance & Auto-Capture

`CameraX • Google MediaPipe • Android SDK`

01 / THE PROBLEM

Solo creators and everyday smartphone users struggle to achieve flattering, well-composed portrait shots without an experienced photographer directing posture, camera tilt, and rule-of-thirds alignment in real time.

02 / EDGE COMPUTER VISION PIPELINE

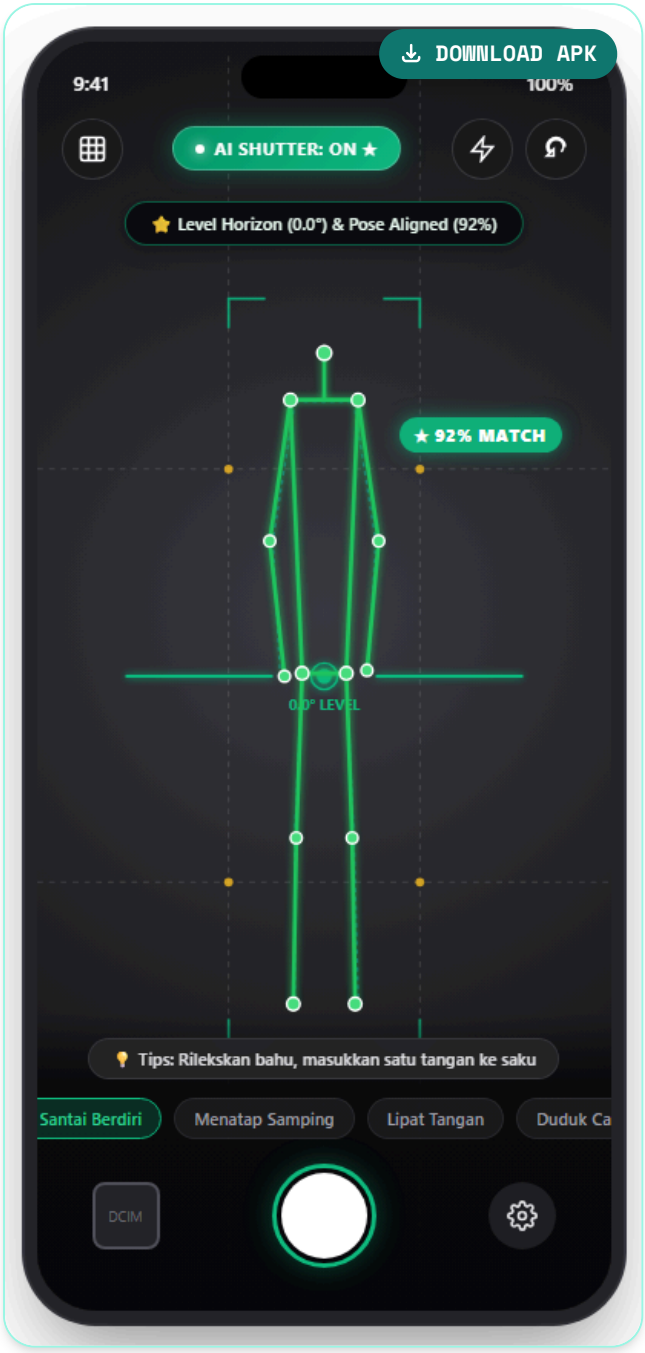
Engineered an Android camera pipeline leveraging **Google MediaPipe Tasks Vision** to track 33 3D skeletal landmarks at 30+ FPS directly on-device. Formulated real-time vector trigonometry calculating joint-angle similarity and Euclidean distances against curated aesthetic pose presets, synchronized with hardware accelerometer sensor fusion for horizon leveling (±1.5° precision).

03 / HANDS-FREE IMPACT & PRIVACY

Implemented the **AI Auto-Shutter** engine: once user posture satisfies preset alignment thresholds (≥ 70% match score) and camera leveling is maintained for 1.2s, the shutter fires automatically without touch. Operates 100% offline on-device with zero cloud roundtrips and absolute user privacy.

> **04 / PERSONAL REFLECTION**
> *"Edge AI is at its best when it disappears into the user's natural workflow. ShutterAI proved that on-device computer vision can democratize artistic photography by transforming mathematical landmark matrices into intuitive creative guidance."*

05 / LIVE APP VISUAL EVIDENCE & EXECUTABLE



↓ Tap image to download ShutterAI-Camera.apk (Android • ~63MB)

CASE 05 • LEADERSHIP & CRISIS OPERATIONS

Logistics & Security Coordination — SMAN 1 Sumedang P5 Exhibition

Resource Management

01 / THE REAL-WORLD CHALLENGE

Tasked with managing logistics, safety, and equipment setup for hundreds of participating students during the flagship school exhibition under severe budget constraints and an unexpected shortfall of external corporate sponsorships.

02 / STRATEGIC ADAPTATION

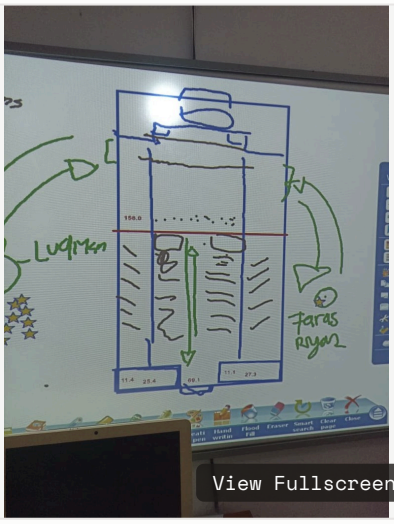
Performed line-by-line cost rationalization, reallocated existing campus gear to prevent unnecessary equipment rentals, and instituted a rotating student security patrol schedule.

03 / OUTCOME & IMPACT

The multi-day cultural and environmental exhibition was completed flawlessly with zero security incidents, zero property loss, and strict adherence to the restructured micro-budget.

> **04 / PERSONAL REFLECTION**
>
> *"Circumstances rarely align with our ideal expectations. Real leadership is not about having endless resources, but about staying composed, adapting rapidly, and solving problems with what is in front of you."*

05 / OPERATIONAL DOCUMENTATION & FIELD EVIDENCE



VENUE LAYOUT & SECURITY FLOW MAPPING

Interactive digital board schematic charting pedestrian movement corridors, booth zoning boundaries, and rotating security patrol checkpoints during the coordinator briefing.



LOGISTICS STAGING & MICRO-BUDGET ALLOCATION

On-ground inventory sorting, ration packages, and consumable distribution organized under restructured micro-budget

parameters to guarantee zero resource deficit during peak exhibition hours.

Honors & Formal Documentation

Authentic Certificates & Citations



GOLD MEDALIST & PROVINCIAL DELEGATE

O2SN Karate Putra – West Java Provincial Championship 2025

Department of Education of West Java Province (Dinas Pendidikan Jawa Barat). Issued July 30, 2025. Certificate No: `16992/PK.03.03.01/PSMA`.

Won 1st Place at the Regency level of Sumedang to earn qualification as the official representative for West Java province. Demonstrates long-term athletic discipline and competitive mental stamina.



ACADEMIC SCIENCE OLYMPIAD

National Science Olympiad (OSN-K) 2025 – Geography

Issued by Balai Pengembangan Talenta Indonesia (BPTI) & SMAN 1 Sumedang on June 26, 2025.

Demonstrated quantitative and spatial reasoning in environmental systems, territorial demographics, and geoscience analysis under strict time constraints.

Competencies & 3-Year Action Plan

Future Trajectory: 2026 – 2029

01 / SOFTWARE & TECH

- **Edge AI & Computer Vision:** Google MediaPipe Tasks, on-device pose estimation, CameraX.
- **Android Kotlin:** Native app development, NFC Host Card Emulation.
- **Backend Services:** Node.js, REST API endpoints, token verification.
- **Foundational Cyber Security:** Authentication protocols, network security principles.

02 / QUANTITATIVE SCIENCE

- **Mathematics & Physics:** Analytical calculus, algebra, logical modeling.
- **Empirical Research:** Qualitative interview design, thematic coding.
- **Financial Analysis:** Cashflow auditing, budget allocation.

03 / CHARACTER & GRIT

- **Karate Discipline:** Poise, calm under duress, respect for code of ethics.
- **Operational Adaptability:** Resourcefulness during critical financial shortfalls.
- **Autodidactic Learning:** Persistent self-study of advanced tech stacks.

{ THE ACTION PLAN / I WILL }

Strategic Roadmap (2026 – 2029)

Year 01 (2026-2027)

Academic Culmination & Screening

Graduate with academic distinction from SMAN 1 Sumedang; complete rigorous screening for the MEXT Undergraduate Scholarship (Japan) and the Institute of Public Administration (IPDN); build foundational cryptography skills.

Year 02 (2027-2028)

Cyber Security & Administrative Systems

Specialize in high-reliability network defense and secure data communications; study public administration principles to align technology with state-level digital sovereignty.

Year 03 (2028-2029)

Systemic Public Contribution

Author applied research on securing governmental public service architectures; design intelligent auditing prototypes to defend public digital infrastructure against systemic cyber threats.

SMAN 1 Sumedang, West Java, Indonesia
linkedin.com/in/aden-nurlaya-4a9611344

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